

Taufiqul Islam Khan

Sessional Lecturer • Research Assistant • MAMBA Lab • University of Manitoba
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CAREER OBJECTIVE

A dedicated educator and researcher with over Five years of combined experience in academia and software engineering. Committed to fostering student success through evidence-based teaching, and to advancing research in software engineering, code review automation, and AI-driven development tools.

WORK EXPERIENCE

University of Manitoba Sessional Instructor

January 2026 – Present
Winnipeg, Manitoba, Canada

- Deliver lectures and lead coursework for COMP 3350 Software Engineering I to undergraduate students
- Instruct students in agile methodologies, software design principles, and industry-standard testing practices
- Provide targeted academic support and constructive feedback to improve student comprehension and performance
- Mentor student teams through full-cycle software projects, and design, administer, and evaluate assessments and examinations
- Manage course content on UMLearn, hold regular office hours, and oversee exam invigilation

BRAC University Full-Time Lecturer

June 2021 – December 2024
Dhaka, Bangladesh

- Served as Course Coordinator for Software Engineering and Introduction to Robotics (Lab), overseeing curriculum delivery and departmental coordination
- Taught undergraduate courses in Software Engineering, Programming Language I, Computer Interfacing, and Introduction to Robotics
- Conducted laboratory sessions and fulfilled departmental administrative responsibilities alongside teaching duties
- Designed examination papers, graded assessments, and invigilated theory and practical examinations

BJIT Ltd. Software Engineer

October 2021 – July 2022
Dhaka, Bangladesh

- Developed and maintained backend services using Spring Boot, REST APIs, and 3dSpace database integrations
- Managed project workflows and version control using Jira, Redmine, Bitbucket, and GitHub in an agile team environment
- Applied software design patterns (Adapter, Observer), wrote unit tests, and performed systematic code refactoring to improve maintainability

ServiceEngine BPO Software Engineer

July 2020 – October 2021
Dhaka, Bangladesh

- Engineered web scraping pipelines using Groovy, Python, and Selenium to extract structured data from HTML sources at scale
- Transformed unstructured data from PDFs and images into structured formats using OCR and image processing techniques
- Built and deployed supervised machine learning models in Python and TensorFlow to validate and quality-check processed data
- Architected cost-efficient automated data processing pipelines using Jenkins CI/CD and AWS cloud infrastructure

Department of CSE, BRAC University Student Tutor

January 2019 – May 2020
Dhaka, Bangladesh

- Supported course instructors in delivering Algorithms and Digital Logic Design to undergraduate students
- Facilitated laboratory sessions, invigilated examinations, and assisted with marking and student feedback

EDUCATION

University of Manitoba M.Sc. in Computer Science (Current Student)

2025 – Ongoing
Winnipeg, Canada

- Research Assistant at MAMBA Lab under Professor Shaowei Wang, focusing on software engineering, code review automation, and LLM-based development tools

BRAC University B.Sc. in Computer Science & Engineering | CGPA: 3.9 / 4.0

2020
Dhaka, Bangladesh

- Thesis: 6LoWPAN Based Futuristic Architecture for Smart Home

Dhaka College H.S.C. in Science | GPA: 5.00 / 5.00

2015
Dhaka, Bangladesh

- Obtained Golden GPA 5 (Highest Score in Bangladesh) and placed 82nd in Dhaka Board

Shere Bangla Nagar Govt. Boys' High School S.S.C. in Science | GPA: 5.00 / 5.00

2013
Dhaka, Bangladesh

- Obtained Golden GPA 5 (Highest Score in Bangladesh)

ACHIEVEMENTS

- Selected in VC’s List and Dean’s List, BRAC University (2016–2020)
- Merit-based Scholarship — Awarded 70% tuition fee waiver, BRAC University (2016–2020)
- Higher Secondary Government Scholarship — Placed 82nd in Dhaka Board, Dhaka College (2015)
- Primary School Government Scholarship — Placed 17th in District, Manikganj, Dhaka (2007)

SKILLS

Frameworks:	Spring Boot, React, Flask
Programming:	Python, Java, HTML, CSS, JavaScript, SQL
Version Control:	Git, Bitbucket
Miscellaneous:	Linux, Shell (Bash), LaTeX (Overleaf), Microsoft Office, MongoDB
Research:	Software Engineering, Data Science, Machine Learning, IoT
Soft Skills:	Proactiveness, Time Management, Teamwork, Problem Solving, Documentation, Engaging Presentation

PROJECTS

Client’s Back-End Development with Spring Boot (BJIT Ltd., 2022)

- Implemented software design patterns including Adapter and Observer to improve system modularity and scalability

Developers Connector with MERN Stack (BJIT Ltd., 2022)

- Built a live full-stack social platform enabling developers to connect, share posts, and collaborate on technical problems

Automated License Plate Detection (ServiceEngine BPO, 2021)

- Developed a real-time computer vision system to detect and recognise license plates from complex road traffic video feeds

Data Analysis and Automated Script Writing with Python (ServiceEngine BPO, 2021)

- Developed automated Python scripts to process large-scale datasets and reformat them according to client-defined specifications

Opu’s Coaching Android App (BRAC University, 2019)

- Designed and built an Android application to connect teachers and students for structured online learning sessions

Leveraging Blockchain and Machine Learning for a Secure and Transparent Voting System (University, 2026)

- Developed a full-stack e-voting platform combining blockchain smart contracts with ML-based facial recognition to deliver secure, tamper-resistant, and privacy-conscious electoral processes with real-time biometric verification

PUBLICATIONS

A Survey of Code Review Benchmarks and Evaluation Practices in Pre-LLM and LLM Era

TI Khan, S Wang, H Zhang, TH Chen • arXiv Preprint • <https://doi.org/10.48550/arXiv.2602.13377> • 2026

6LoWPAN Based Futuristic Smart Architecture for Home Automation

SMJ Hasan, M Ali, TI Khan, F Afrin, MM Islam • ICAICT 2020 • DOI: [10.1109/ICAICT51780.2020.9333483](https://doi.org/10.1109/ICAICT51780.2020.9333483) • Cited by: 3

Transfer Learning Using Convolutional Neural Networks for COVID-19 Detection with X-Ray Images

PK Das, M Rashidunnabi, TI Khan, SRR Ahmad • scholar.google.ca/citations?user=yINPGNUAAAAJ

LANGUAGES

Bangla:	Native Proficiency	English:	Professional Proficiency IELTS: 6.5
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REFEREES

Shaowei Wang

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